

INDRANEEL GHOSH

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WORK EXPERIENCE

Software Engineer

Amazon Web Services

Jul 2025 – Curr

Seattle, USA

- Leading multiple **performance, resiliency and monitoring initiatives for the AWS Ultraservers fleet**. We operate a rapidly growing production fleet of 6000 NVL72(360k GPUs) racks consisting of GB200s and GB300s.
- Driving a **multi-product Re:Invent initiative** to build **Ultraserver monitoring and anomaly detection systems** that can help us detect quality and performance issues faster. This also impacts **EKS and Slurm Ultraserver offerings**.
- Developed a **stress testing infrastructure** used for cluster qualification and performance benchmarking. The infrastructure also enables teams to **run cross-rack workloads** to reproduce customer issues and **benchmark training and inference performance for various open-source models**.
- Drove improvements in the hardware qualification process to improve delivery time without compromising fleet quality. **Reduced qualification time by 25%** by driving infrastructure and code-level optimizations.
- Drove initiatives to improve repair automations. This improved the average delivery speed from **5 to 50 new NVL72 racks per week**. Also, **reduced node-level repair SLAs by 80% and rack-level SLAs by 20%**. We honor **SLAs for GPU replacement and rack-level repairs of 4 and 42 hours, respectively**. **Increased ARR per rack by 32%**.
- Redesigned **EC2's quality monitoring systems to measure the Ultraserver fleet's annual failure rate accurately**. Drove adoption of these systems to a production fleet of **25k accelerator racks**.

Software Engineer III

Google

Sept 2024 – Jul 2025

Hyderabad, India

- **Technical lead for GKE Cluster Director**. Worked with cross functional stakeholders to drive initiatives that helped improve GCP's ClusterMax rating from **Bronze to Silver in a semester**.
- Developed **workload-agnostic AIP compliant APIs for provisioning and managing clusters** for **Cluster Director**. Defined and implemented the **regional expansion plan for the product**.
- **Contributed to KEPs for Topology Aware Scheduling in Kueue** to support the **GKE performance goals for A3+ cluster families**. These launches **drove USD 250 million+ annual revenue growth**.
- Built GKE cluster test bed to run scaling tests for new machine families and **drove helm adoption in cluster provisioning systems** to reduce provisioning failures. Provisioned GKE clusters with **peak scale of 65k nodes**.
- Drove initiative to simplify open source LLM fine-tuning on GKE using **hyperparameter optimised GPU and TPU recipes to reduce development time by 85%**. Built a ray-based platform that enabled engineers to **benchmark and design custom training and inference recipes**.
- Implemented tooling to **simplify resource sharing and management between projects**. Improved **developer productivity for 10 teams** and saved **59 million USD** in annual infrastructure costs.

Software Development Engineer II

Amazon

Jan 2021 – Sept 2024

Bangalore, India

- Re-architected **Prime Video's payment systems to move to a configuration-based architecture to reduce time required to launch new payment methods by 35%**. Worked with multiple teams to align their migration plans, which **led to a YOY revenue increase of 220 million USD**.
- Worked on adding **support for installment-based payment methods** in Prime Video Commerce Systems. The program led to an increase in ARR of **150 Mn+ USD**.
- Built **training and inference primitives** to serve **content and ad recommendations for miniTV**. Inference systems handled peak TPS of **8000**, training systems handled 2TB of data spanning across text, image and video files. Leveraged **FastAPI, Guice and Sagemaker** for inference and **Step Functions, Glue and Pyspark** for training pipelines. **Increased average streaming time by 18% and ad revenue by 13%**.
- Drove **migration of inference systems from Flask to FastAPI** to reduce model serving costs by 40%. Upstreamed the changes to a framework **used by more than 300 teams, leading to xxx million USD in annual cost savings**.
- Led multiple infra optimisation initiatives for Prime Video and miniTV. **Migrated systems to AWS offerings with lower TCO and fixed architectural inefficiencies across multiple systems**.

ACHIEVEMENTS

- **World Rank 825** in Kickstart Round H 2020.
- **Leetcode Best Contest Rating** of 2125.(Top 1% on the platform. Awarded Guardian Badge)

EDUCATION

Birla Institute of Technology and Science, Pilani

B.E Computer Science (CGPA: 8.5)

Class of 2021

Pilani, Rajasthan, India